





▶ We can measure total communication between referent and target using the index

► A **restart** has occurred on machine/replica m if the index increases from 0 to N

Restart timer indicates when a machine propagates a new reference sample to the target

► **A round trip** has occurred on machine/replica m if the index increases from 0 to N to 0

► Let $T_{0,-}^m$ be the first time the m -th replica is at the reference

$$T_{0,-}^m = \inf\{t : (I_t^m, \epsilon_t^m) = (0, -1)\}$$

- The i -th restart on machine m occurs at time $T_{i,+}^m$

$$T_{i,+}^m = \inf\{t > T_{i-1,-}^m : (I_t^m, \epsilon_t^m) = (N, 1)\},$$

► The i -th round trip machine m occurs at time $T_{i,-}^m$

$$T_{i,-}^m = \inf\{t > T_{i,+}^m : (I_t^m, \epsilon_t^m) = (0, -1)\}$$

► A replica with an index $I_f^n = N$ indicates that machine n is contributing to the target



Round trips indicate when a machine undergoes a refreshment

► A replica with index $I_f^n = 0$ indicates a reference has been generated at time t

ROUND TRIPS AND RESTARTS

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- ▶ A replica with index $I_t^m = 0$ indicates a reference has been generated at time t
- ▶ Round trips indicate when a machine undergoes a refreshment
- ▶ A **round trip** has occurred on machine/replica m if the index increases from 0 to N to 0
 - ▶ The i -th round trip machine m occurs at time $T_{i,-}^m$

$$T_{i,-}^m = \inf\{t > T_{i,+}^m : (I_t^m, \epsilon_t^m) = (0, -1)\}$$

- ▶ A replica with an index $I_t^m = N$ indicates that machine m is contributing to the target
- ▶ Restart times indicate when a machine propagates a new reference sample to the target

ROUND TRIPS AND RESTARTS